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The Impact of Different Genres of Music on our Health

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The Notes (Data-Centric Project)
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Blog Post URL: *https://medium.com/@jsubramanian_50302/the-impact-of-different-genres-of-music-on-our-health-199f776461a6*

Personal Motivation

The impact of music on stress levels is a widely discussed topic, but little quantitative research has been done on how different genres affect physical stress markers. Beyond just the data-driven analysis and research we have conducted, the root of our motivation and interest stems from our passion and love for different styles of music. Being out-of-state students, we have faced different forms of adversity trying to fit into college and develop a routine and life away from home — we view music as an escape from our day-to-day lives. While music brings us together, we have found that we have very diverse music tastes, which we wish to quantify the impacts of and assess on our health. We hope to further the current state of music-based research while finding results that have a personal impact on our daily lives.

Background and Research Problem

Our project assesses how listening to different genres of music impacts various health metrics. We are extremely interested in evaluating the

intertwining of health and music, hoping to gain further insight through personal analysis.

The specific questions we hope to answer are as follows:

1. **How do different genres of music impact the heart rate/$SpO_2$of an individual?**
2. **Does a preference for a genre result in unique stress-related physiological symptoms?**
3. **Does the level of preference for a music genre correlate with lower baseline stress levels before an activity?**

To answer our individual questions, the specific health metrics we are analyzing are **SpO_2**, **Heart Rate**, and **Overall Stress Levels**.

SpO_2 and **Heart Rate** are **semi-automatic data sources**; we are using a pulse oximeter for this collection and then manually plugging in the daily data onto a master spreadsheet (CSV). On the other hand, **Overall Stress Levels** is a **manually collected** data source in which we have created a **6-point scale** with specific descriptions and levels of stress for each point in the scale. We have normalized the scale to ensure that our interpretations of stress are similar and do not create high levels of variance in our data collection.

Stress Level Scale:

1. **Typical Daily Stress** — Seen as the daily level of stress we would have on an average day, including going to class, doing some work, and taking care of our daily routine and lives. This is manageable and does not impact our stress and mood.
2. **Moderate Stress** — Seen as a level of stress that requires slightly more attention, such as a deadline coming up, or just general work that must be done promptly. Still, there is no major impact on stress or mood, but there is a need to complete something, which changes a person's rate of work. You can function well, but start to feel slight pressure.
3. **Noticeable Stress** — Seen as stress that begins to slightly impact mood and concentration. There might be slight anxiety, along with a general challenge to relax.

4. **High Stress** — Seen as stress that fully impacts mood, concentration, and day quality. You may find it hard to relax in these scenarios and may begin to lose sleep, etc. This may also be the beginning of paranoia.
5. **Extreme Stress** — Stress level that significantly impacts the capability of a person to function daily. There is an impairment of judgment and focus, which can come with anxiety and extreme fear.
6. **Panic Attack Level Critical Stress** — Comparable to the stress that could be felt an hour prior to a major final exam, which you have not prepared for. At this point, your body function may begin to shut down, and you are unable to properly handle any tasks, etc. You may feel completely overwhelmed and feel sweating, loss of breath, etc.

In terms of data collection, we are collecting our SpO_2 levels in the morning, after we wake up, and before going to bed. We intend to assess whether listening to a specific type of music throughout the day makes a difference to this metric. Concerning stress level and heart rate assessments, we take these measurements before and after doing a daily routine activity. We all have chosen an activity that we do on average for at least 1–2 hours per day (i.e, eating lunch, doing work, practicing LeetCode) and plan to center our music listening around this time. We want to see how our overall health metrics change while doing this activity, which also allows us to take conceptual notes on our performance doing these tasks with certain genres of music. Additionally, we have 6 pre-determined genres of music and Spotify playlists that we will be rotating through: **Jazz, Shoegaze, Rap, Carnatic and Western Classical, Trap,** and **Indie**. We have also ranked our preference order of these 6 genres of music to account for preference bias impacts. This is the premise of when and how we plan to collect data.

Solution

Answering the Research Questions

Collecting the data sources listed above helps us answer our research questions, as it helps break down specific health metrics that are directly correlated to certain behaviors we are hoping to learn more about. Our questions are generally oriented around explicit impacts on the heart rate, SpO_2 levels, and overall stress levels of an individual. We are directly measuring 3 of these metrics, while also getting insight into preference bias impacts by assessing baseline stress levels for each individual. Additionally,

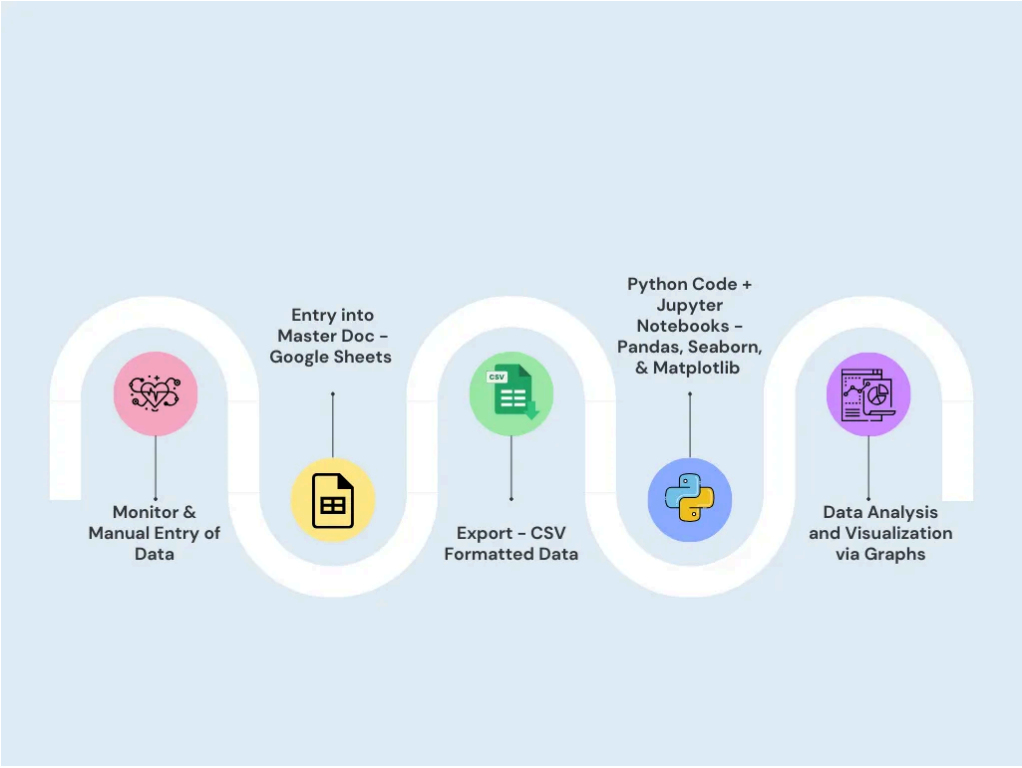
factors such as stress can impact metrics such as heart rate, which are being collected within our data collection and conceptual notes.

Elaborating from a music standpoint, we are noting the genres of interest that we are listening to, which gives a basis from which our research questions can be answered. Our questions are generally looking at the impacts of music from a generalized genre and tempo perspective, which are all aspects that can be sourced from knowing just the genre itself. Many of these genres and the playlists we built have uniform tempo and style of music within them, which helps form generalizations. Many genres of music have natural tendencies in terms of tempo, instruments used, and overall level of sound. With these generalizations kept in mind, we can source the appropriate information we need. This helps answer how certain factors within each style of music impact our health metrics.

Limitations

With regard to providing accurate answers to our research questions, our data collection approach is limited by a **small sample size**, **personal music bias**, and **negligible results in SpO_2 level changes**. The data we collected is restricted to 3 individuals, which is not enough data and information to formulate comprehensive results. While we may answer our questions on a personal basis, these insights may not apply to a larger audience or grouping of people. Additionally, we all have certain styles of music that we are naturally affiliated with, which makes getting objective data challenging when answering our questions. While we account for preference bias, the data can still be skewed to favor certain genres. From a SpO_2 metric standpoint, we identified that there was a negligible change in SpO_2 levels throughout the study, which makes this data invalid. While we formulated insights based on this metric, it did not show meaningful results from a quantitative standpoint. On a generalized note, we are very different individuals, and the presence or absence of music can have varied impacts upon us; this can be challenging to manage when attempting to answer our research questions. While we do not believe these aspects of our data collection will become major issues, these are factors that must be acknowledged.

Data Pipeline



Data Pipeline Visualization

Tools and Overall Process of Pipeline

All data was entered into a **standardized Google Sheets** table daily by participants. The sheet includes: **date, morning/night SpO_2, HR before/after activity, stress before/after, music genre listened to, and preference of the genre ranked**. The data was then exported as a **CSV file**, which was imported into **Jupyter Notebooks using Pandas** (Python Language Base). Pre-processing involved converting the date format using datetime, and calculating 3 new columns: **HR Change = HR After — HR Before, Stress Change = Stress After — Stress Before, SpO_2Change = Night SpO_2 — Morning SpO_2**. This data was then cleaned and formatted in visualizations using **Seaborn, Matplotlib, and Plotly**.

Challenges Faced and Changes

A challenge we faced in earlier milestones was the **influence of preference bias** on certain health metrics and our overall data. To account for this, we had to change our initial CSV spreadsheet to include a **preference ranking column**. This column would rank from 1 to 6 for each individual, their favorite to least favorite genre of music. This factor helped us change some of our research questions to focus on the impacts of music preference on certain health metrics and subsequent behaviors of each individual. The

changes made had primarily successful outcomes, as we were able to identify certain trends and insights of how preferences impact health metrics such as heart rate. At the same time, this inclusion of data did not provide quantifiable evidence of a correlation between preferences and baseline stress. Overall, making these additional changes had a **net-positive impact** that allowed our group to account for further variability in the data.

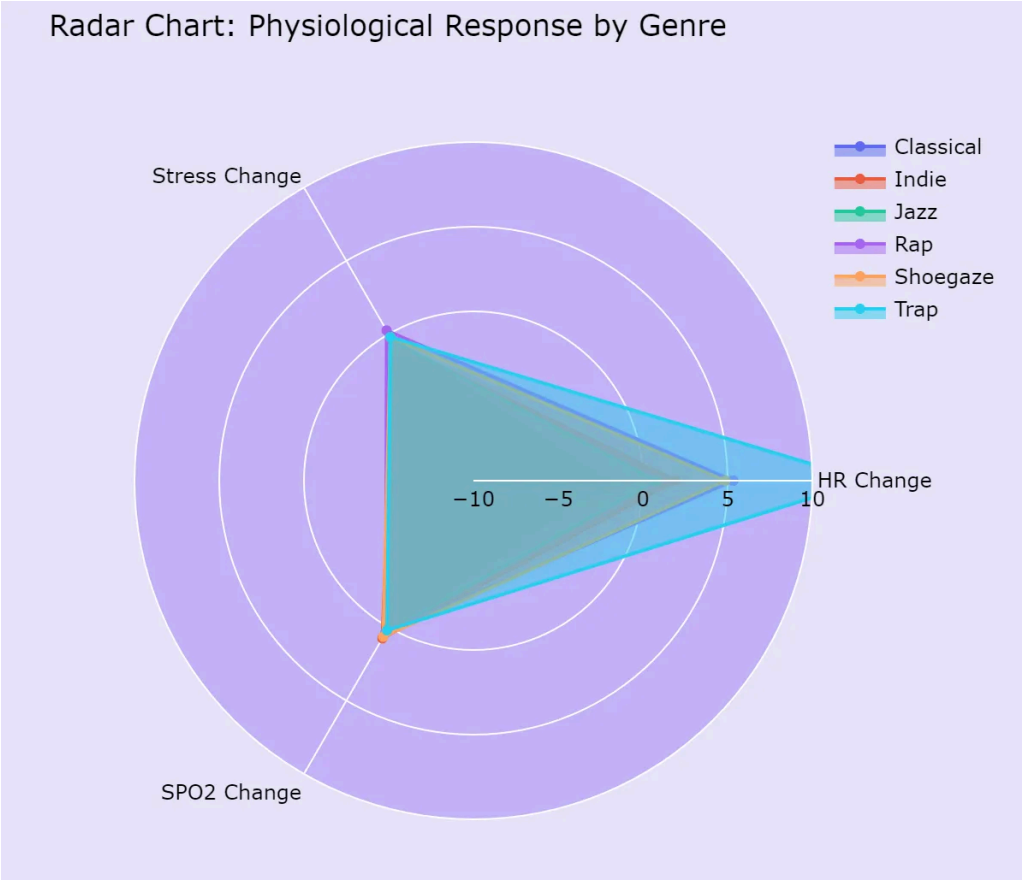
Insights

Intermediate Analyses & Thought Process

To begin answering our individual research questions, we first calculated the following metrics — **HR Change, Stress Change, and SPO2 Change** — derived from before/after activity data. Initially, we had visualized genre effects across all users using **simple bar, violin, and 3D scatter plots**. While informative, these had combined all our data and did not necessarily reveal our individual trends that we expected based on our hypotheses. Additionally, we introduced a "Preference" column to our data (1= most preferred, 6 = least preferred). This addition allowed us to visualize how users' affinity to a genre might shape physiological change.

Radar Chart (Genre Profiles)

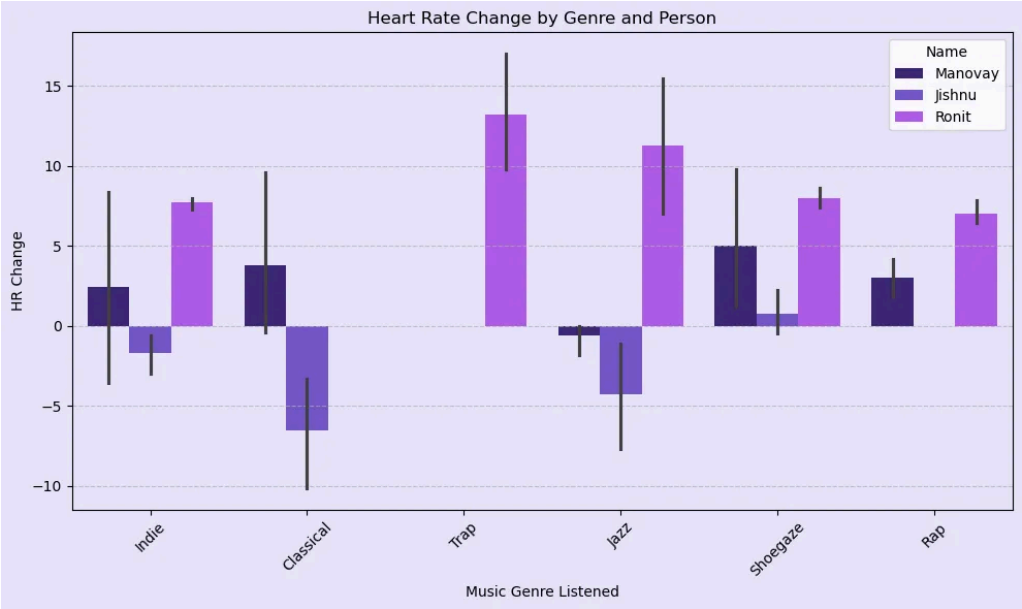
To express the **physiological effects of each genre**, we created a **radar chart**, where each axis represents one of the tracked metrics:



Classical and Jazz had some of the lowest HR and stress changes, visually shrinking their radar shapes, showing that relaxing genres improve physiological regulation. **Indie and Shoegaze** showed **balanced profiles**, while **Trap** showed the **strongest HR spike**. This plot is thus able to express the "shape" of each genre's physiological footprint on our subjects.

Bar plot (Heart Rate Change by Genre and Person)

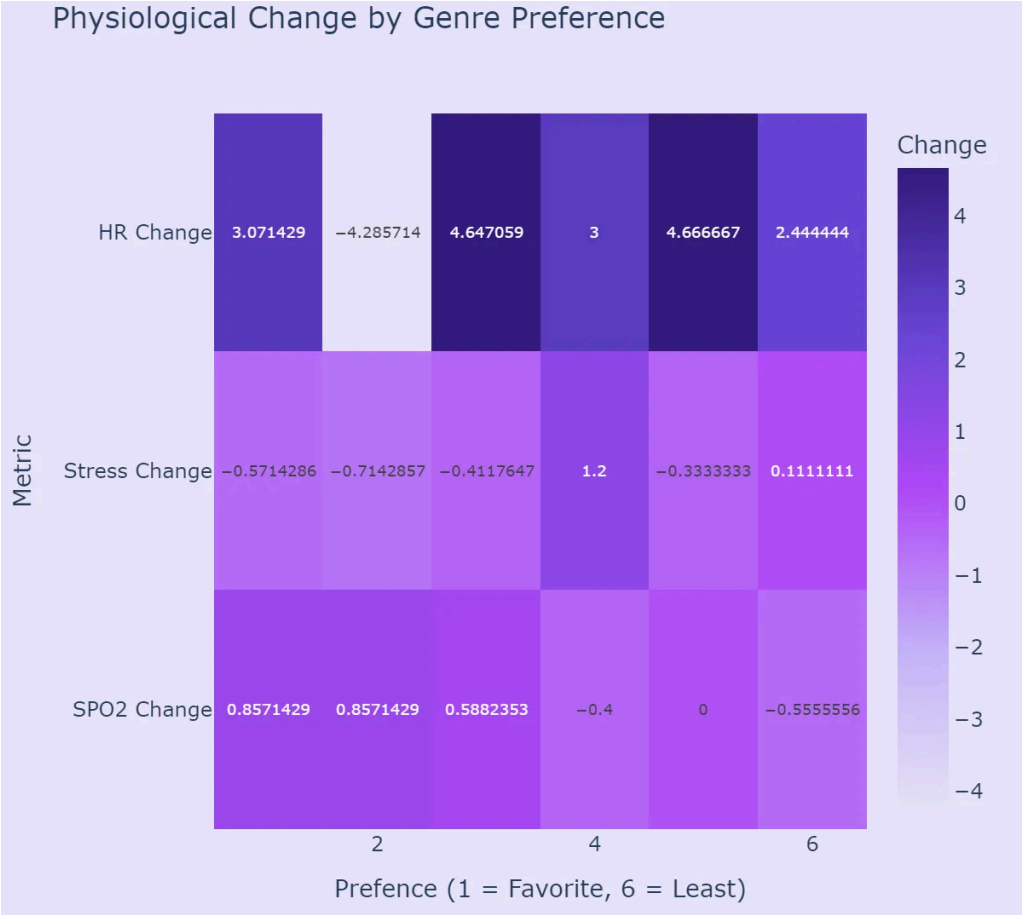
This **bar chart** shows the **heart rate change by music genre per individual.** Ronit's question focused on whether different genres can physically stimulate or calm the listener, and the goal of this visualization is to answer that question.



From this visual, we can see that **Ronit experiences significant HR raises when listening to Trap or Rap**, suggesting that these genres have an energizing effect on him. In contrast, **Jazz music calms down Jishnu**, indicating a calming physiological effect. This variability demonstrates how **genre choice can meaningfully impact physical stress markers**.

Heatmap (Preferences vs Change)

To address Manovay's question on — Does a preference for a genre result in unique stress-related physiological symptoms? — We created a **heatmap** that **aggregates the average physiological change per preference level.** Each cell in the heatmap shows how genres, rated at a given preference level, impact their user:

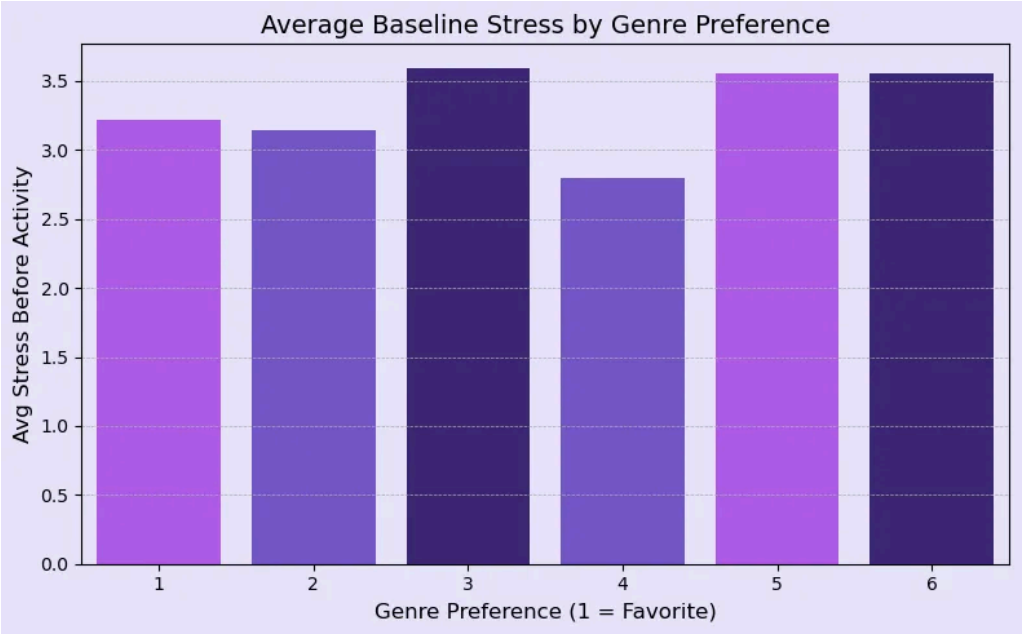


From this visualization, we can see that a **strong negative stress change is linked with preference level**, as the **favorites seem to result in larger stress decreases.** The least preferred genres show positive Stress Change, indicating reduced effectiveness and maybe stimulating effects. HR shows variability throughout the preferences, and SPO2 is the least sensitive to preference, as it shows subtle fluctuations; thus, nothing significant can be discerned for either.

This directly supports Manovay's question that individuals experience peak stress reduction with genres they prefer. It also quantifies how "bias" or preference isn't just subjective, as it has measurable physiological effects.

Bar Chart (Baseline Stress vs Genre Preference)

This **bar chart** aims to examine Jishnu's question on whether genre preference correlates with lower baseline stress before an activity.



However, the **results are inconclusive**, there is no clear trend as **stress levels fluctuate across preference rankings**, with a low at 4 and a spike at 3. This lack of consistency makes us believe that **genre preference might not be a reliable predictor of baseline stress levels**. This visualization is thus lacking, but we believe that if we collected more data, we could discover a trend.

Challenges

A key challenge was **splitting our data into individual variability**, as the same genre produced inconsistent results across users. Adding the 'Name' and 'Preference' columns helped to clarify this. Another challenge was the **lack of value of the SPO2 metric**. SPO2 remained in the same 95–99 zone, and the fluctuations were very minute, rendering an attribute of our data essentially useless. **Visualization clarity** was an issue, so we tweaked the color palettes until we found something that is able to easily communicate data across.

Questions

We were able to answer our questions with the data given when we introduced the preference and individual tracking aspects. For instance, the heatmap revealed that higher preference levels were associated with greater stress reduction, lending support to Manovay's idea that individuals were calmed when listening to music they enjoyed. Additionally, the genre-specific analyses showed that stimulating genres like Trap and Rap elevated Ronit's heart rate, which helped answer his question on how genres affect heart rate specifically. However, our attempt to answer Jishnu's question on

baseline stress in relation to genre preference was not conclusive, as no trend emerged.

Tracking Experiences

1. Ronit: My experience throughout the tracking process was generally a positive experience. At first, I felt that the process of collecting data multiple times a day would get frustrating, however, I soon realized after a few days of repeating it, the process becomes much easier once you become accustomed to it. A highlight of my tracking experience was when I realized that when I was listening to one of my favorite bands, my heart rate elevated, which was very interesting to see how enjoyment towards music can be seen on a physical level. One positive outcome I had during my tracking experience was when I realized how high my heart rate was on the day before my exam, which made me sit down and relax, which in turn helped lower my heart rate and allow me to focus more during the exam. One negative outcome I however had was realizing how high my heart rate gets when I am frustrated, which is something I plan to work on in the future.

2. Manovay: My expectations were simple and my experience validated that. I knew that if I just set alarms and followed a routine I would be consistent with my readings. It helps that the readings only took a few minutes and fit well into my schedule, without need for additional work or focus. A positive outcome came as I had wrongly believed that my resting heart rate would be pretty high since my family has a history of heart/cholesterol problems, but I was relieved to see that the heart rate was usually pretty low, even when I was engaged in strenuous tasks (LeetCode). A negative outcome was just noticing how stressed I often was and what those periods were, such as midterm weeks. This project stuck out to me as the data I collected personally connected to me and my friends. I was excited to tinker with it and see the insights on my friend and I's subconscious behaviors. In all, the experience has been greatly valuable and I have learnt a lot of technical and soft skills from it.

3. Jishnu: In hindsight, my expectations from the tracking experience were to get a comprehensive understanding of how music impacts my daily life and overall health in a positive and/or negative manner. I wanted to better my health and viewed music as a gateway to some form of self-betterment, which was the root of my expectations from this tracking experience. Regardless of the results, I hoped this tracking experience would allow me to create a routine of listening and applying music to

better my health, which was vital. I think the highlight of my tracking experience was being given the opportunity to analyze in real-time how certain genres of music impact my health metrics. Even if I had taken subconscious notes of how I feel when listening to certain genres of music, I never fully took the time to analyze such impacts; this was a highlight and a great experience I took from this. Additionally, being able to see a deliverable and analysis developed based on my own data was very worthwhile and created a personal connection and drive to what we were doing as a group. Being able to take a hobby and a very important aspect of our lives and create it into something meaningful was a great highlight of this entire experience. A positive outcome I sourced from this experience would be my understanding of what conditions suit me best when doing work, including the style and genre of music that allows me to focus well and stay relaxed. I was able to see a lot of connections between my heart rate, sleep quality, and the genre of music I listened to throughout the day, which will help me in the long term create a healthier routine. Additionally, I now know that classical styles of music, including Carnatic music and Jazz, are genres that I am naturally affiliated with, which help me find a sense of emotional and physical peace. Additionally, I found out that I become hyper-fixated and focused when listening to EDM-style music, which was something I did not expect to learn about myself. While this style of music was not an extensive part of the study, I did take time to assess this genre individually due to the influence of this experiment. Through this experience, I was able to identify such insights that will help better my health. While this outcome was not fully negative, I did see that many of the days within my tracking experience, I was facing concerning levels of stress beyond levels 2 and 3 on the stress scale. While I do not view this as a full negative outcome, it is something I must consider when creating a routine for myself that is both healthy and productive. I am constantly seeing high levels of stress based on my self-inputted data, which may be detrimental to my overall health in the long term. “Burn-out” was an issue I faced, and this data made it clear that while music helped lower such stress levels, I need to find a better balance within my daily life. Overall, this experience has been worthwhile and something I will cherish and appreciate for a very long time. Being able to gain insights into the intricacies of music while also being able to work with my good friends made this experience extremely rewarding.

Contribution Statements

1. Ronit: Within Milestone 1, I mainly focused on building the technical side of the project, focusing on the data streams and analysis sections. For Milestone 2, I helped with the pre-processing of the data, working with the dataframes to help prepare it for initial visualizations. Finally, for Milestone 3, I assisted with building the new visualizations of our data.
2. Manovay: For Milestone 2, I worked with Ronit to build the technical components of our project, developing the data streams and analysis sections of the deliverable. Within Milestone 2, I developed the basis of the technical code of the visualizations and helped with initial analysis. For the final Milestone, I spearheaded the data visualization section and overall analysis of the graphs in relation to our new research questions.
3. Jishnu: Within Milestone 1, I developed the project definition and goals portion, building the conceptual side of the project. Moving onto Milestone 2, I planned out and refined the data pipeline, and built the overall CSV tracker and other resources that were used for data collection. Finally, in Milestone 3, I assisted with the data analysis and insights while also spearheading the blog post development.

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